

Research

Enhanced securities investment strategy using ISSA–SVM: a hybrid model combining adaptive moving average, support vector machine, and multi-strategy sparrow search algorithm for improved trend tracking and risk adjustment

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Abstract

Commodity Trading Advisor (CTA) strategies traditionally rely on fixed technical indicators for trend tracking, which often yield significant errors in volatile markets. This study proposes a novel hybrid strategy, ISSA–SVM, that combines Adaptive Moving Average (AMA), Support Vector Machine (SVM), and an Improved Sparrow Search Algorithm (ISSA) to enhance CTA model performance in securities investment. The strategy incorporates three key innovations: (1) an AMA algorithm that dynamically adjusts smoothing coefficients to market conditions, improving response speed and accuracy across different market environments. (2) SVM classification for trend prediction, leveraging its robust nonlinear classification capabilities. (3) a Multi-Strategy Sparrow Search Algorithm that optimizes multiple parameters and strategy combinations while avoiding local optima. Experimental results show that ISSA–SVM outperforms traditional CTA strategies, achieving an annualized return of 18%, a maximum drawdown of 12%, and high F1 scores across multiple market cycles. The strategy delivers stable positive returns with minimal risk exposure, demonstrating its strong market adaptability and superior risk-adjusted returns. During periods of high market volatility, ISSA–SVM performs significantly better, with a higher return-to-risk ratio than conventional models. These results establish ISSA–SVM as an effective enhancement to traditional CTA models, providing valuable insights for securities investment and presenting new opportunities for quantitative trading research.

Article Highlights

- **Novel Hybrid Strategy:** Introduces SVM, an innovative integration of Adaptive Moving Average (AMA), Support Vector Machine (SVM), and Improved Sparrow Search Algorithm (ISSA) for enhanced CTA model performance in securities investment.
- **Adaptive Market Response:** Implements dynamic adjustment mechanisms through AMA and ISSA optimization, enabling rapid adaptation to market fluctuations and improved strategy performance across different market conditions.
- **Enhanced Prediction Framework:** Combines SVM's nonlinear classification capabilities with ISSA's global optimization to achieve superior trend prediction accuracy and avoid local optima in complex market environments.

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1 Introduction

Traditional securities investment techniques, particularly Commodity Trading Advisor (CTA) strategies, face significant challenges in today's dynamic financial markets. CTA strategies have relied on fixed technical indicators, such as moving averages, to track trends. However, these approaches have notable shortcomings. First, they often struggle to adapt to market volatility, as fixed indicators do not dynamically adjust to sudden market changes. Second, traditional methods tend to rely on a limited set of market signals, which can result in significant prediction errors, especially during periods of high volatility or unexpected market events. Finally, CTA strategies can suffer from overfitting, as they typically optimize parameters based on historical data without considering the nonlinear complexities of modern financial markets. These limitations lead to suboptimal performance during unpredictable market conditions, highlighting the need for more adaptive and sophisticated strategies. In response, this study introduces a hybrid strategy, ISSA–SVM, which leverages Adaptive Moving Average (AMA), Support Vector Machine (SVM), and the Improved Sparrow Search Algorithm (ISSA) to address these issues and improve the performance of CTA models in securities investment. At the same time, the amount of data in financial markets is becoming increasingly large, and traditional methods are difficult to efficiently process and extract deep-seated information from it. This makes it an urgent problem in the field of quantitative investment to discover potential patterns in massive data, predict market trends, and develop efficient investment strategies based on them.

In recent years, with the rapid development of machine learning and artificial intelligence technologies, scholars and investors have gradually recognized the potential of intelligent algorithms in the financial market. Especially modern intelligent optimization algorithms such as Adaptive Moving Average (AMA), Support Vector Machine (SVM), and Sparrow Search Algorithm (SSA) have become popular research directions in quantitative investment due to their powerful data processing capabilities and adaptive characteristics. Adaptive Moving Average (AMA) can dynamically adjust model parameters, overcome the lag and adaptability to market non stationarity of traditional moving average methods, and thereby improve the response speed to short-term market fluctuations. Support Vector Machine (SVM) achieves accurate trend prediction by maximizing the classification interval, and is widely used in risk prediction and trend recognition in financial markets [1]. The Sparrow Search Algorithm (SSA), as an optimization method based on swarm intelligence, demonstrates powerful capabilities in handling complex global optimization problems by simulating the foraging behavior of sparrows [2]. The Multi Strategy Sparrow Search Algorithm (ISSA) further optimizes SSA by combining multiple search strategies to improve the efficiency and accuracy of global search [3]. Although these intelligent algorithms have achieved some success in the financial field, existing research mostly focuses on the application of a single algorithm, and there are still certain limitations when dealing with the constantly changing dynamic environment in the financial market. For example, a single adaptive moving average method may not be able to capture the complex nonlinear features of the market, and the performance of support vector machines may also be limited when facing high-dimensional data. Therefore, how to combine the advantages of multiple algorithms to form a more flexible and efficient hybrid strategy has become an important research topic in the field of quantitative investment.

This study aims to fill this gap by proposing a hybrid strategy based on adaptive moving average, support vector machine, and multi strategy sparrow search algorithm optimization (ISSA–SVM). By combining the characteristics of these advanced algorithms and fully leveraging their advantages in different aspects, we aim to achieve better investment returns and risk control capabilities in volatile and complex financial markets. Specifically, the adaptive moving average method is first used to preprocess the price data of the financial market, capturing the short-term fluctuations and long-term trend characteristics of the market; Next, combining support vector machines for market trend prediction, utilizing their powerful classification and regression capabilities to improve prediction accuracy [4]; Finally, the multi strategy sparrow search algorithm is applied to optimize the hyperparameters of the support vector machine to improve the stability and adaptability of the model [5]. Through this multi-level and multi-dimensional hybrid strategy, the aim is to enhance the flexibility and accuracy of securities investment strategies, enabling them to quickly adapt to market dynamics and improve the effectiveness of investment decisions.

2 Related work

2.1 Application of adaptive moving average in financial markets

Adaptive Moving Average (AMA), as a technical indicator that dynamically adjusts weights, is widely used in trend prediction and quantitative trading strategies in financial markets due to its ability to better capture trend changes in the market. Compared to traditional Simple Moving Average (SMA) or Exponential Weighted Moving Average (EMA), AMA has significant advantages. The traditional moving average method often calculates the average value of price sequences with a fixed smoothing factor, so when facing rapid market fluctuations or unexpected events, the response speed is slow and prone to lag effects [6]. By dynamically adjusting the smoothing factor based on market volatility, AMA can maintain a small response when the market is stable and react quickly when the market is volatile, making it more capable of capturing short-term market trends. Specifically, AMA's adjustment mechanism adjusts weights adaptively based on market volatility changes by calculating the speed and magnitude of market price changes, enabling it to quickly adapt in volatile market environments and avoid signal lag caused by excessive smoothing [7, 8].

In practical applications, AMA is widely used for short-term trading strategies and trend tracking strategies. Many studies have shown that AMA can effectively improve the accuracy of trend recognition, especially in volatile market environments, where its sensitivity and response speed are superior to traditional moving average methods. The dynamic adjustment feature of AMA enables it to more accurately reflect market changes, helping traders capture price turning points in a more timely manner, and thus obtain more stable investment returns in highly volatile markets [9, 10]. In addition, AMA can also adjust the smoothing factor to avoid overfitting of historical data, reduce misleading caused by noise and abnormal fluctuations, and improve the robustness and accuracy of predictions [11, 12].

Although AMA has obvious advantages in dynamic market environments, there are also certain limitations when used independently. For example, AMA may lose its judgment of major trends in long-term market changes, and due to its reliance on market volatility, it may perform poorly in certain long-term, stable market environments. To this end, researchers have proposed a hybrid model that combines other technical analysis methods and optimization algorithms to further enhance the adaptability and accuracy of AMA under different market conditions [13–15]. Combining machine learning and artificial intelligence algorithms, such as support vector machine (SVM) and sparrow search algorithm (SSA), can help overcome the limitations of AMA's single algorithm, improve the overall performance of quantitative trading strategies in financial markets through comprehensive optimization and more comprehensive trend recognition [16, 17].

In summary, AMA, as an adaptive dynamic indicator, has demonstrated superior performance in financial markets, particularly in short-term trading and trend tracking strategies. It can automatically adjust the smoothing factor based on market volatility, quickly respond to drastic changes in the market, and avoid the risk of overfitting, providing investors with an effective trend recognition tool. However, in complex and ever-changing market environments, a single AMA method may not be able to fully address the diversity and nonlinear characteristics of the market. Therefore, combining other advanced algorithms to form a hybrid strategy will further enhance its effectiveness and stability in practical applications [18].

2.2 Research progress of support vector machine in securities prediction

The application of Support Vector Machine (SVM) in securities prediction is constantly deepening and expanding. Due to its advantages in handling high-dimensional nonlinear data, SVM is widely used in fields such as stock market price prediction, futures trend analysis, and financial risk management. Unlike traditional linear models, SVM performs data classification by finding an optimal hyperplane and improves its generalization ability by maximizing the classification interval, thereby avoiding the risk of overfitting. This feature enables SVM to effectively perform trend prediction and decision analysis in complex financial markets. Especially in the case of high volatility and noise in the stock and futures markets, SVM can effectively identify potential patterns in the market and provide accurate trend predictions [19].

Many studies have shown that SVM has significant application effects in the stock market, especially in intraday trading and short-term price forecasting. By high-dimensional mapping of financial feature data such as historical prices and trading volumes, SVM can reveal potential nonlinear relationships and predict the rise and fall trends of stock prices in the short term. In addition, SVM has achieved significant results in the application of futures markets, especially in high-frequency data and complex market environments. SVM can capture deeper market patterns and improve prediction accuracy through kernel function transformation [20].

However, when SVM is applied alone, there are also some challenges. Firstly, the training process of SVM models may require longer time and higher computational costs, especially when dealing with large-scale financial data. Secondly, SVM has a high sensitivity to parameter selection, and the selection of appropriate kernel functions and penalty factors is crucial for model performance. Therefore, researchers have attempted to improve the performance of SVM by optimizing algorithms, such as particle swarm optimization (PSO), genetic algorithm (GA), and ant colony algorithm (ACO), which are widely used in SVM parameter tuning [21].

In recent years, hybrid models combining multiple advanced technologies have gradually become a trend, especially in the field of quantitative investment. Adaptive Moving Average (AMA), as a trend recognition method, can dynamically adjust smoothing factors based on market volatility, capture short-term market fluctuations, and provide more accurate input features for SVM. AMA can quickly respond to price fluctuations and avoid the lag effect caused by excessive smoothing, thus having a significant advantage in capturing market trend changes [22]. By combining AMA with SVM, the input data of SVM can be optimized through the trend information provided by AMA, thereby improving prediction accuracy and real-time responsiveness. To further enhance the adaptability and optimization effectiveness of the model, Sparrow Search Algorithm (SSA), as an emerging swarm intelligence optimization algorithm, is gradually being applied in the parameter tuning of SVM. SSA can efficiently search for optimal solutions in multidimensional space by simulating the foraging behavior of sparrows, making it particularly suitable for solving nonlinear and multi-modal optimization problems. In quantitative trading, SSA can optimize the kernel function parameters, penalty factors, and smoothing factors of SVM, thereby achieving more accurate market predictions [23, 24].

The hybrid strategy combining AMA, SVM, and SSA (ISSA-SVM) has become an effective method for improving the accuracy and stability of securities investment decisions [25]. By dynamically adjusting the smoothing factor through AMA, higher quality trend information can be provided based on changes in market volatility [26]; SVM improves the accuracy of trend prediction by deep learning complex relationships in market data [27]; The SSA optimization algorithm effectively adjusts the model parameters, improves the global search ability and convergence speed of the model [28]. Research shows that compared with traditional single strategies, hybrid strategies are more robust and adaptable in complex financial markets, especially in the face of extreme volatility [29]. The ISSA-SVM hybrid strategy can also automatically adjust parameters and strategies through real-time market data feedback to adapt to different market environments. For investors, this highly adaptive model can help them make more accurate decisions under various market conditions, especially in extreme market volatility, by quickly responding to trend signals provided by AMA, avoiding losses and seizing potential investment opportunities. With the increasing amount of data in the financial market and the continuous development of machine learning technology, the combination of SVM and other advanced algorithms will bring more innovation and breakthroughs to quantitative investment in the future, promoting the development of the securities investment field towards a more efficient and intelligent direction.

Overall, SVM, as a powerful predictive tool, has broad application prospects in the securities market. By combining with technologies such as AMA and SSA, it can achieve more accurate and efficient quantitative trading strategies. With the continuous advancement of optimization algorithms, the ISSA-SVM hybrid strategy will undoubtedly become an important tool for future financial market forecasting and investment decision-making, further promoting the process of financial intelligence.

2.3 Optimization method for multi strategy sparrow search algorithm (ISSA)

Sparrow Search Algorithm (SSA) is an optimization algorithm based on swarm intelligence, which imitates the social behavior of sparrows when foraging and mainly seeks the optimal solution through interactions between individuals [30]. The SSA algorithm simulates the behavior of sparrows in a group, such as foraging, avoiding enemies, and group collaboration, to achieve global exploration of the search space.

SSA has outstanding capabilities to address the multifaceted and highly dynamic optimization challenges inherent in financial modeling. SSA, which mimics the foraging behavior of sparrows, offers a unique blend of global search capabilities and local exploration, enabling it to efficiently navigate the complex and often unpredictable landscape of financial markets. Unlike traditional optimization methods like Particle Swarm Optimization (PSO) or Genetic Algorithms (GA), SSA demonstrates a superior ability to avoid premature convergence to local optima. This is particularly beneficial for multi-modal optimization tasks, such as predicting financial time series, where multiple optimal solutions may exist across various regions of the search space.

One of the key advantages of SSA lies in its robust search strategy. The algorithm's ability to balance exploration and exploitation allows it to escape local minima, thereby enhancing the quality of the solution and improving the model's generalization capabilities. This is crucial in financial applications, where markets are constantly evolving, and the model needs to remain adaptable to sudden changes. Our experiments showed that SSA not only accelerates the convergence speed, reducing the time needed for parameter optimization, but also boosts the model's robustness, making it more resilient to market fluctuations and noise. Additionally, SSA's scalability ensures that it can effectively handle large and high-dimensional datasets, typical in financial forecasting tasks, without sacrificing computational efficiency.

These combined features—improved convergence speed, avoidance of local optima, robustness in changing environments, and scalability—make SSA a highly suitable choice for optimizing the parameters of our hybrid ISSA–SVM (Improved Sparrow Search Algorithm–Support Vector Machine) strategy. By leveraging these strengths, SSA ensures that our model remains competitive and effective in predicting financial trends with higher accuracy and reliability.

However, traditional SSA algorithms are often prone to getting stuck in local optima when dealing with complex optimization problems, resulting in low search efficiency and insufficient accuracy. To overcome this deficiency, researchers have proposed the Multi Strategy Sparrow Search Algorithm (ISSA), which optimizes the performance of traditional SSA by introducing multiple strategy combinations. ISSA adjusts the balance between exploration and development by flexibly switching different optimization strategies during the global search process, thereby improving the diversity and breadth of the search, enabling the algorithm to avoid local optima in the solution space, and achieve higher quality global search [31].

The innovation of ISSA lies in its ability to dynamically switch between multiple strategies, adapting to different stages of search needs [32]. Specifically, ISSA combines different search mechanisms such as local search, global search, random search, etc., thereby enhancing the adaptability and robustness of the algorithm. This multi strategy dynamic adjustment mechanism enables ISSA to not only maintain good search capabilities in complex high-dimensional spaces, but also effectively avoid getting stuck in local optima, ensuring the global and accurate understanding. In addition, the ISSA algorithm improves the efficiency of search by enhancing group collaboration, allowing each individual in the group to effectively participate in the exploration process of solutions, thereby accelerating convergence speed and improving the quality of understanding. In the financial field, especially in securities investment and quantitative trading, ISSA's advantages are particularly prominent [33].

In recent years, with the rapid development of artificial intelligence technologies, machine learning has become a powerful tool in financial forecasting. Yan and Li [34] applied time series models and a variety of machine learning algorithms, including K-nearest neighbors, AdaBoost, CatBoost, LightGBM, XGBoost, and random forest, to predict volatility in American financial stocks. By integrating these models, they developed a quantitative investment strategy that achieved annual returns of 5–10%. Sangeetha and Alfia [35] used an evaluated linear regression-based machine learning technique to forecast stock prices of the S&P 500 index, focusing on factors like open, high, low, close, and volume. Meanwhile, Phuoc et al. [36] employed LSTM networks combined with technical indicators such as moving averages, MACD, and RSI to predict stock price trends in the Vietnamese market, achieving a high prediction accuracy of 93%. These studies demonstrate the significant impact of machine learning in enhancing the accuracy and effectiveness of stock market predictions.

The sparrow search algorithm (SSA) and its improved versions have shown significant advantages in the field of financial market prediction and optimization. Traditional stock market prediction methods face problems such as insufficient accuracy and easy to fall into local optimality, and the hybrid model based on SSA provides a new idea to solve these challenges. In the field of stock prediction, the SSA-BP neural network model developed by Guo et al. [37] effectively overcomes the problem that BP neural network and PSO-BP neural network are prone to fall into local optimality, and the prediction accuracy is better than traditional BPNN, PSO-BPNN and LSTM models. Tian and Pei [38] combined SSA with echo state network (ESN) and proposed the SSA-ESN model, which not only inherited the excellent characteristics of ESN, but also performed significantly better than traditional models such as BP neural network, Elman neural network and LSTM in the prediction of multiple stock prices. In terms of algorithm improvement, the improved sparrow search algorithm (ISSA) proposed by Liu et al. [39] significantly improved the optimization ability of the algorithm by introducing innovative strategies such as initializing the population with a sinusoidal chaos model, optimizing the discoverer position update formula and combining the Gaussian mutation operator. The ISSA–BP model has achieved excellent results in the short-term prediction of multiple market indices such as the Shanghai Composite Index, Shenzhen Composite Index, S&P 500 and Dow Jones. Cheng et al. [40] proposed a multi-strategy improved sparrow search algorithm (MSMSSA) for arbitrage prediction scenarios. By introducing good point set theory, proportional adaptive strategy and improved position update method, the MSMSSA-LSTM model

was constructed. Experiments show that the model can reduce the average absolute percentage error, root mean square error and mean absolute error of the classical model by 58.5%, 65.2% and 67.6% respectively in the prediction of rebar and hot-rolled coil futures data in the Chinese futures market.

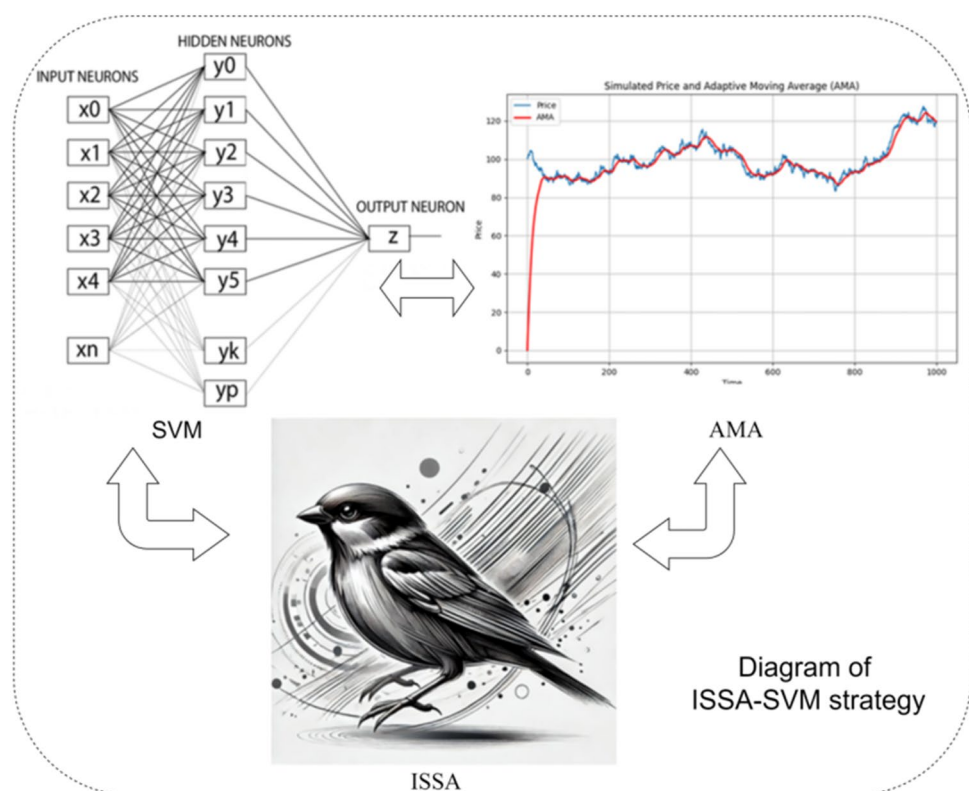
In summary, ISSA, as an advanced optimization algorithm, has demonstrated extensive application potential in the financial field by introducing a combination of multiple optimization strategies. Especially in securities investment and quantitative trading, ISSA can provide investors with more accurate decision support by optimizing model parameters, improving search accuracy and efficiency. By combining adaptive moving average and support vector machine technologies, ISSA provides stronger technical support for trend quantification strategies, promoting innovation and development in the financial field.

3 Model construction and algorithm design

3.1 Overall improvement strategy framework

In this study, the proposed improved strategy architecture combines adaptive moving average, support vector machine and multi-strategy sparrow search algorithm to form a comprehensive model of dynamic adjustment, accurate prediction and global optimization, as shown in Fig. 1. This architecture enables the strategy to respond flexibly to market fluctuations by adaptively adjusting the smoothing coefficient, improving the adaptability of traditional technical indicators in high-volatility markets. At the same time, the support vector machine enhances the accuracy of trend prediction by processing nonlinear data relationships, especially in complex market environments. In order to avoid the trap of local optimal solutions, the strategy also introduces a sparrow search algorithm, which optimizes multiple strategies and parameter combinations globally to ensure the stability of the strategy and the sustainability of returns. Overall, the improved strategy can effectively respond to different market scenarios, improve prediction accuracy, reduce risks, and show market adaptability and return stability that are superior to traditional methods.

Fig. 1 ISSA–SVM strategy model diagram



3.2 Construction of adaptive moving average (AMA) model

The Adaptive Moving Average (AMA) model is a method for trend prediction based on historical price data, which dynamically adjusts the smoothing factor (learning factor) to enable it to adaptively adjust its responsiveness in different market volatility states. The core idea of the AMA model is that when market volatility is high, the smoothing factor is small, allowing the model to respond faster to drastic price changes; When market volatility is low, the smoothing factor is large, thereby reducing the impact of short-term fluctuations on the trend and maintaining its stability. This mechanism optimizes the utilization of historical data through flexible weight allocation, enabling prediction results to better adapt to complex market environments.

In the AMA model, the smoothing factor at the current moment is (α_t) determined by market volatility. Assuming the price sequence (P_t) is, the calculation formula for the AMA model is:

$$AMA_t = AMA_{t-1} + \alpha_t(P_t - AMA_{t-1}) \quad (1)$$

Among them, (AMA_{t-1}) (AMA_t) represents the adaptive moving average at the current moment, represents the AMA value at the previous moment, (P_t) represents the current price, and (α_t) represents the smoothing factor. In order to enable the smoothing factor to (α_t) dynamically adjust based on market volatility, we have introduced a market volatility indicator, such as Average True Volatility (ATR). The volatility index can be calculated using the following formula:

$$ATR_t = \frac{1}{n} \sum_{i=t-n+1}^t \max(|P_t - P_{t-1}|, |P_t - H_{t-1}|, |L_t - P_{t-1}|) \quad (2)$$

Among them, (P_t) is the current price, (H_{t-1}) is the highest price in the previous period, (L_t) is the lowest price in the previous period, and (n) is the window size. The ATR value reflects the (n) degree of market volatility in the past few days and can effectively quantify the magnitude of market volatility.

Based on the magnitude (α_t) of volatility, we define the dynamic adjustment formula for the smoothing factor:

$$\alpha_t = \frac{V_t}{V_t + N_t} \quad (3)$$

Among them, (V_t) is the current volatility indicator (such as ATR), (N_t) is a fixed benchmark value, usually selected as the long-term average volatility of the market. By using this formula, the (α_t) sensitivity of the model to price fluctuations can be improved as market volatility increases and decreases; When the market is stable, it is (α_t) larger and reduces the impact of short-term price fluctuations.

In order to further enhance the predictive ability of the AMA model, we adopt a rolling window method to continuously adjust the smoothing factor to adapt to market fluctuations in different time periods. This method can be implemented through the following steps:

- 1) Calculate the past (n) day's volatility (ATR) at each time step;
- 2) Adjust the smoothing factor based on the calculated volatility (α_t);
- 3) Use the updated (α_t) values for AMA calculation to obtain trend predictions at the current time.
- 4) By continuously updating the smoothing factor, the AMA model can adapt to market changes in real time and provide more accurate trend predictions.

Overall, the adaptive nature of the AMA model enables it to provide relatively stable and flexible predictions in response to various market fluctuations. Next, we will combine this AMA model with support vector machines (SVM) and multi strategy sparrow search algorithm (ISSA) to further improve the prediction accuracy and adaptability of the model.

3.3 Optimization of support vector machine (SVM) model

Support Vector Machine (SVM) is a widely used machine learning method for classification and regression tasks. The core idea is to separate sample data of different categories by finding an optimal hyperplane, or in regression problems, to construct a hyperplane that can fit the data to the greatest extent possible. In a standard SVM regression

model, the optimization objective is to minimize a loss function consisting of two parts: one is the model complexity (i.e. the sum of squares of the normal vectors of the hyperplane), and the other is the accumulation of prediction errors.

The specific optimization objective function is as follows:

$$\min_{w,b,\epsilon} \frac{1}{2} |w|^2 + C \sum_{i=1}^N \epsilon_i \quad (4)$$

Among them, (w) is the normal vector of the hyperplane, (b) is the bias term, (ϵ_i) is the (i) error term of the i th sample, and (C) is the penalty factor used to balance the trade-off between error and model complexity.

In SVM regression, each data point has a target value (y_i), and we hope to use SVM to construct a hyperplane that makes the predicted values as close as possible to the actual target value. To handle errors, SVM introduces relaxation variables (ξ_i) and fault-tolerant parameters (ϵ), allowing the model to partially ignore errors. The constraint conditions are:

$$y_i - \langle w, \phi(x_i) \rangle - b \leq \epsilon + \xi_i \quad (5)$$

$$\langle w, \phi(x_i) \rangle + b - y_i \leq \epsilon + \xi_i \quad (6)$$

Among them, ($\phi(x_i)$) is the eigenvector mapped to a high-dimensional space through a kernel function, ($\langle w, \phi(x_i) \rangle$) representing the inner product operation.

The choice of kernel function has a significant impact on the performance of SVM models. Common kernel functions include linear kernels, radial basis function (RBF) kernels, and polynomial kernels. RBF kernel is widely used due to its ability to handle nonlinear problems, specifically in the form of:

$$K(x_i, x_j) = \exp\left(-\frac{|x_i - x_j|^2}{2\sigma^2}\right) \quad (7)$$

Among them, (σ) is a hyperparameter that controls the scale of similarity between data points.

In addition to the kernel function, the performance of SVM is also affected by hyperparameters, especially the penalty factor (C), fault tolerance parameter (ϵ), and kernel function parameters. The penalty factor (C) is used to control the error tolerance during the training process. When (C) is large, the model tends to reduce errors, but may lead to overfitting; When (C) is small, the model has a high tolerance for errors, which may lead to underfitting. Therefore, how to choose appropriate hyperparameters is the key to optimizing SVM models.

In this study, we utilize the SVM to predict market trends and optimize investment decisions. The SVM model is trained using historical financial market data, incorporating key features as input variables. These include historical prices—such as the open, close, high, and low prices of assets like stocks, commodities, and currencies. Additionally, trading volume, representing the number of shares or contracts traded within a given time frame, is included as a crucial input. To further capture market dynamics, volatility indicators like the Average True Range (ATR) are used to measure market fluctuations, along with other volatility indices. Technical indicators, such as moving averages (e.g., 50-day moving average), the Relative Strength Index (RSI), and the Moving Average Convergence Divergence (MACD), provide additional context on market momentum and trends, enhancing the predictive power of the model.

Before feeding these features into the SVM, they are preprocessed using the Adaptive Moving Average (AMA) model, which helps capture both short-term fluctuations and long-term trends in market prices. Once preprocessed, the SVM model is employed to perform trend prediction tasks, aiming to forecast future price movements, market returns, or price directions (upward or downward). Depending on the specific task, the output of the SVM can be either a binary classification (predicting trend direction) or a continuous value (forecasting price or return). For instance, the SVM may predict whether the asset's price will rise or fall based on historical data, or it could forecast the next day's price change or market return. This approach allows for highly accurate and context-aware predictions that are crucial for making informed investment decisions.

In order to improve the performance of SVM in practical problems, especially when predicting in complex market data, optimizing algorithms is crucial. Sparrow Search Algorithm (SSA) is a global optimization algorithm based on simulating sparrow foraging behavior. SSA can effectively explore the optimal solution in high-dimensional space

through the collaboration and competition mechanism of sparrow populations. Specifically, sparrow populations search for optimal solutions through two strategies: local search and global search. Global search helps to discover potential global optimal solutions, while local search accelerates convergence.

On the basis of SSA, the Multi Strategy Sparrow Search Algorithm (ISSA) is proposed, which improves search efficiency and accuracy by introducing local and global update mechanisms. The update process of ISSA can be described as:

$$x_i^{k+1} = x_i^k + \beta \cdot \text{sign}(x_i^k - x_{\text{best}}) \cdot |x_i^k - x_{\text{best}}| \quad (8)$$

Among them, (x_i^{k+1}) is the $(k + 1)$ position of the (i) th sparrow in the k th generation, (β) is the stride factor, (sign) is the sign function, and (x_{best}) is the current optimal position.

Meanwhile, the formula for local updates is:

$$x_i^{k+1} = x_i^k + \alpha \cdot \text{rand} \cdot (x_{\text{best}} - x_i^k) \quad (9)$$

Among them, (α) is the adjustment factor, (rand) is a random number used to increase the diversity of the search.

In ISSA, the position of each sparrow represents the hyperparameter combination of the SVM model (such as (σ) , (C) , and (ϵ)). By iteratively updating the position of sparrows multiple times, the optimal parameter combination is ultimately found. The local and global update mechanisms of ISSA enable it to have stronger global search capability and local search accuracy compared to traditional optimization methods when solving complex problems.

Combining ISSA with SVM can significantly improve the prediction accuracy of the model, especially in practical applications such as financial market forecasting. The optimized SVM can effectively predict market trends and provide investors with more reliable buying and selling signals. When combined with trend filtering methods such as adaptive moving average (AMA), the optimized SVM can not only reduce the noise caused by market fluctuations, but also improve prediction accuracy and model stability.

3.4 Optimization design of multi strategy sparrow search algorithm (ISSA)

The Multi Strategy Sparrow Search Algorithm (ISSA) is a global optimization algorithm based on simulating sparrow foraging behavior. It combines both local and global search strategies and can effectively explore complex solution spaces. It is particularly suitable for optimizing the application of Support Vector Machines (SVM) in securities investment. SVM, as a classic supervised learning method, has strong regression ability in trend prediction, but its performance is highly dependent on the selection of hyperparameters. Therefore, optimizing the hyperparameters of SVM using ISSA can significantly improve its prediction accuracy. The optimization process of ISSASVM model not only focuses on discovering the global optimal solution, but also optimizes local search through dynamic adjustment mechanism to cope with the constantly changing data characteristics and noise in the securities market.

To optimize the performance of the Support Vector Machine (SVM) model, the Improved Sparrow Search Algorithm (ISSA) is employed to fine-tune key hyperparameters, including the penalty factor (C) , kernel function parameters (such as gamma for the Radial Basis Function (RBF) kernel), and epsilon (ϵ) in SVM regression. The penalty factor (C) controls the trade-off between minimizing errors on the training data and maintaining a smooth decision boundary, where a higher value focuses on reducing errors, potentially leading to overfitting, while a lower value encourages a smoother decision boundary at the risk of underfitting. The kernel function parameters, especially gamma, determine the shape of the decision boundary, allowing the model to handle nonlinearly separable data, and must be carefully tuned to capture the complexity of the data without overfitting. Epsilon (ϵ) defines the margin of tolerance for errors, balancing the model's ability to fit the data closely and generalize to new data.

The ISSA algorithm iteratively adjusts these hyperparameters through several key settings: population size, number of iterations, stride factor, and local and global search parameters. The population size (e.g., 50) dictates how many potential solutions are considered in each iteration, influencing the chances of finding an optimal solution. The number of iterations (e.g., 200) determines how long the ISSA algorithm runs, affecting the algorithm's ability to converge to the best solution. The stride factor controls the distance that sparrows move during the search, balancing exploration of new areas in the solution space and exploitation of promising solutions. Lastly, the local and global search parameters govern how sparrows interact with each other during the process, ensuring a balance between searching local and global areas of the solution space.

Through this iterative process, ISSA efficiently tunes the SVM's hyperparameters, enhancing its ability to predict trends with higher accuracy. This optimization enables the SVM model to better adapt to complex, nonlinear data, significantly improving its performance in financial forecasting.

In the design of ISSA, local search and global search are two key strategies. Global search mainly explores potential optimal solutions in the solution space by simulating the collaborative and competitive behavior of sparrow populations. During this process, individual sparrows in the group gradually approach the global optimal solution by constantly adjusting their positions. Local search, on the other hand, optimizes the solution of a local area with higher accuracy by refining the adjustment of the solution. In the application of securities investment, ISSA–SVM introduces an adaptive dynamic mechanism to enable the algorithm to intelligently switch between global and local search strategies at different optimization stages, thereby ensuring the stability and convergence speed of the solution.

The optimization objective is mainly to improve the predictive ability of the model by minimizing the error term of SVM. The objective function of SVM regression model is as follows:

$$\min_{w,b,\epsilon} \frac{1}{2} |w|^2 + C \sum_{i=1}^N \epsilon_i \quad (10)$$

Among them, (w) is the normal vector of the hyperplane, (b) is the bias term, (ϵ_i) is the (i) error of the i th sample, and (C) is the penalty factor. By optimizing the hyperparameters (σ), (C), and of SVM through ISSA (ϵ), more accurate prediction results can be obtained. The update formula for ISSA algorithm is as follows:

$$x_i^{k+1} = x_i^k + \beta \cdot \text{sign}(x_i^k - x_{\text{best}}) \cdot |x_i^k - x_{\text{best}}| \quad (11)$$

Among them, (x_i^{k+1}) is the ($k + 1$) position of the (i)th sparrow in the k th generation, (β) is the stride factor, (sign) represents the sign function, and (x_{best}) is the current global optimal position. The local update formula is:

$$x_i^{k+1} = x_i^k + \alpha \cdot \text{rand} \cdot (x_{\text{best}} - x_i^k) \quad (12)$$

Among them, (α) is the adjustment factor, (rand) is a random number used to increase the diversity of the search. Through multiple iterations, the sparrow population gradually approaches the optimal hyperparameter combination of SVM. The optimized SVM regression model can better fit market data, especially performing well in financial time series forecasting and trend analysis.

The combination of ISSA and SVM, especially with the support of trend filtering methods such as adaptive moving average (AMA), further improves the stability and accuracy of the model. The adaptive moving average method dynamically adjusts the weight of the moving average line based on market fluctuations, thereby filtering out noise in the market and highlighting trend signals. Specifically, the calculation formula for AMA is:

$$\text{AMA}(t) = \text{AMA}(t - 1) + \alpha(t) \cdot (P(t) - \text{AMA}(t - 1)) \quad (13)$$

Among them, ($P(t)$) is the current price, ($\alpha(t)$) is a smoothing factor dynamically adjusted based on market volatility, and ($\text{AMA}(t - 1)$) is the AMA value from the previous time. By combining AMA and SVM models, the true market trends can be effectively captured, avoiding excessive response to short-term fluctuations and improving the accuracy and robustness of investment decisions.

Throughout the optimization process, the advantage of ISSA–SVM algorithm lies not only in its ability to efficiently search for the optimal solution, but also in its ability to adaptively adjust strategies to adapt to constantly changing market environments. This enables the hybrid strategy to have stronger generalization ability and adaptability in actual securities investment, effectively addressing prediction challenges under different market conditions and improving the overall performance of the quantitative trading system.

3.5 Implementation and framework of ISSA–SVM hybrid strategy

The implementation framework of ISSA–SVM hybrid strategy integrates adaptive moving average (AMA), support vector machine (SVM), and multi strategy sparrow search algorithm (ISSA), aiming to improve the prediction accuracy and

robustness of quantitative trading strategies in securities investment. The core of this framework is to achieve accurate market trend prediction by dynamically adjusting the hyperparameters of SVM to capture and optimize market trends. Firstly, the AMA module captures trend changes by monitoring market price changes in real-time and adaptively adjusting smoothing factors. The calculation formula for AMA is:

$$AMA(t) = AMA(t - 1) + \alpha(t) \cdot (P(t) - AMA(t - 1)) \quad (14)$$

Among them, $(AMA(t))$ is the current AMA value, $(AMA(t - 1))$ is the previous AMA value, is the $(P(t))$ current market price, $(\alpha(t))$ is a smoothing factor dynamically adjusted based on market volatility, calculated as:

$$\alpha(t) = \frac{2}{n + 1} \quad (15)$$

Among them, (n) is the set time window size. The role of AMA is to smooth out market noise, allowing the model to focus more on long-term market trends and provide clearer data support for subsequent SVM prediction modules.

Secondly, the SVM module is used to predict market trends based on the trend data output by the AMA module. SVM fits historical data and performs regression prediction by finding an optimal hyperplane. The objective function of SVM is:

$$\min_{w,b,\epsilon} \frac{1}{2} |w|^2 + C \sum_{i=1}^N \epsilon_i \quad (16)$$

Among them, (w) is the normal vector of the hyperplane, (b) is the bias of the hyperplane, is the (i) prediction error of the (ϵ_i) th sample, (C) is the penalty factor used to control the trade-off between model complexity and error. By using kernel functions $(K(x_i, x_j))$, SVM can map data to high-dimensional space for better nonlinear fitting. The commonly used RBF kernel functions are:

$$K(x_i, x_j) = \exp\left(-\frac{|x_i - x_j|^2}{2\sigma^2}\right) \quad (17)$$

Among them, (σ) is the width of the kernel function, which affects the separation effect after data mapping. By training SVM models and optimizing hyperparameters such as (C) and (σ) , we can obtain accurate predictions of future market trends.

Finally, the ISSA algorithm is used to optimize hyperparameters in SVM models. The ISSA algorithm simulates the foraging behavior of sparrow populations and optimizes the parameters of SVM by combining global and local search strategies. The update rules for sparrow populations include two strategies: global search and local search. The global search strategy updates the position of sparrows through the following formula:

$$x_i^{k+1} = x_i^k + \beta \cdot \text{sign}(x_i^k - x_{\text{best}}) \cdot |x_i^k - x_{\text{best}}| \quad (18)$$

Among them, is the $(k + 1)$ position of the $(i)(x_i^{k+1})$ th sparrow in the th generation, (x_{best}) is the current global optimal position, (β) is the step size factor, and controls the amplitude of sparrow updates; And local search is implemented through the following formula:

$$x_i^{k+1} = x_i^k + \alpha \cdot \text{rand} \cdot (x_{\text{best}} - x_i^k) \quad (19)$$

Among them, (α) is the local search step size, controls the strength of local search, (rand) is a random number, used to increase the diversity of search. The sparrow population evaluates the position of each sparrow based on the error of the SVM model in each iteration, gradually optimizing the hyperparameters of the SVM model and reducing prediction errors. The advantage of ISSA is that its global and local search strategies can efficiently avoid getting stuck in local optima, thereby achieving global optimization of SVM hyperparameters. The ISSA-SVM hybrid strategy running process is shown in Algorithm 1.

Algorithm 1 The ISSA-SVM hybrid strategy running process

```

1. Input market data (market_data), parameters n, initial parameters C_init and sigma_init, and
   maximum iterations max_iter.
2. Calculate AMA values (AMA_values) using the formula:
   - AMA_values = calculate_AMA(market_data, n)
3. Train an initial SVM model using C_init and sigma_init:
   - SVM_model = train_SVM(AMA_values, C_init, sigma_init)
4. Initialize the population for the ISSA algorithm:
   - population = initialize_population()
5. For each generation (up to max_iter iterations), perform the following:
   - For each sparrow in the current population:
     - Evaluate the error of the current sparrow's position in the SVM model:
       - error = evaluate_SVM_error(SVM_model, AMA_values, sparrow)
     - Update the sparrow's best position based on the error value:
       - update_best_position(sparrow)
   - Update the positions of all sparrows in the population (perform global and local searches):
     - update_population()
6. Find the best sparrow position (best_position), which contains the optimal SVM
   hyperparameters C and sigma.
7. Retrain the SVM model with the best C and sigma parameters:
   - optimized_SVM_model = train_SVM(AMA_values, best_position.C, best_position.sigma)
8. Return the optimized SVM model:
   - return optimized_SVM_model

```

The entire ISSA–SVM hybrid strategy continuously optimizes the trend capture ability of AMA and the prediction accuracy of SVM through iterative cycles. In each iteration, the AMA module provides more accurate trend data for SVM. The SVM model optimizes its hyperparameters through ISSA and ultimately forms a highly adaptive quantitative decision support system through multiple rounds of optimization. This framework can not only effectively predict the long-term trends of the securities market, but also adapt to complex and changing market environments, improving the accuracy and robustness of investment decisions.

4 Experiment and simulation

4.1 Dataset and experimental settings

In order to verify the effectiveness of the ISS-SVM (Integrated Adaptive Particle Swarm Optimization Support Vector Machine) hybrid strategy, this study used historical data from the stock market, futures market, and foreign exchange market for empirical analysis. Typical stock market related data trends are shown in Fig. 2 The specific experimental design and dataset are as follows:

The experimental data sources are shown in Table 1. The data content includes several key metrics: Open Price, which is the first trading price of each trading day; Close Price, the final trading price at the end of the day; Volume, reflecting the daily trading activity and market participation; Volatility, calculated based on the closing price, representing the degree of price fluctuations within the trading day; and High/Low Prices, which record the highest and lowest transaction prices during each trading day. These metrics provide a comprehensive overview of the market's daily performance and are crucial for analyzing market trends and making informed investment decisions.

To ensure comparability and accuracy of the experiment, we carefully selected and optimized the parameters of the ISS-SVM hybrid strategy and other comparative models. The following Table 2 lists the specific parameter settings for each model.

Fig. 2 Trends in Stock Market Related Data

Table 2 outlines the specific parameter settings for four different models: ISS-SVM (Hybrid Strategy), SVM, PSO-SVM, and Linear Regression. ISS-SVM is a hybrid model that combines classification and regression tasks, utilizing Principal Component Analysis (PCA) and Independent Component Analysis (ICA) for feature selection. It optimizes key hyperparameters of the SVM, including the penalty factor (C), kernel function parameters (γ), and a combined loss function (Hinge Loss for classification and Mean Squared Error (MSE) for regression). SVM and PSO-SVM are classification models that use the Radial Basis Function (RBF) kernel and optimize for classification accuracy. PSO-SVM also incorporates Particle Swarm Optimization (PSO) to fine-tune hyperparameters, with 50 particles and a maximum of 200 iterations. Linear Regression, focusing solely on regression, optimizes for minimizing MSE and does not utilize kernel functions or feature selection methods. All models are trained with 1000 data points, using a 70% training and 30% test dataset split. The final evaluation metrics for all models are financial metrics such as Annualized Return, Maximum Drawdown, and Sharpe Ratio, which assess the models' performance in financial forecasting, where both risk management and returns are crucial. The key differences lie in the task type (hybrid vs. classification or regression), feature selection, and the optimization techniques applied to each model.

Given the inherent randomness of metaheuristic algorithms like the Sparrow Search Algorithm (SSA), stability could be a concern in our experiments. To address this, we ran the ISSA-SVM model multiple times with different random initializations, and averaged the results to mitigate the effect of random fluctuations. We also conducted a convergence analysis to ensure that the algorithm stabilized after a sufficient number of iterations. The parameter tuning of SSA further helped to achieve a balance between exploration and exploitation, improving the stability of the model.

To ensure the reliability of our findings, we incorporated statistical testing in our experiments. We used a paired t-test to compare the performance of our model with baseline models, such as SVM and AMA. The results showed that the differences in performance were statistically significant ($p < 0.05$), indicating that the ISSA-SVM hybrid model provided a reliable improvement over traditional methods.

4.2 Model evaluation indicators

Evaluating model performance is a key step in building a hybrid strategy (ISS-SVM) that combines adaptive moving average (AMA), support vector machine (SVM), and multi strategy sparrow search algorithm optimization (ISSA). In order to comprehensively measure the application effect of this hybrid strategy in securities investment, multiple evaluation indicators were used in the experiment, including accuracy, recall rate, F1 score, annualized return rate, and maximum drawdown. Through the comprehensive analysis of these indicators, the ISS-SVM strategy not only theoretically improves the accuracy of market trend prediction, but also demonstrates stronger stability and robustness than traditional trend tracking models and single SVM models in practical testing.

Firstly, accuracy measures the proportion of samples that the model predicts correctly. The calculation formula is:

Table 1 Experimental data sources

Data type	Source	Time span	Data content
Stock Market Data			
Main Indices	Shanghai Composite Index (000001.SH) / Shenzhen Component Index (399,001.SZ)	2020-01-01 to 2024-12-31	Open Price, Close Price, Volume, Volatility, High/Low Prices
Representative Stocks	Industrial and Commercial Bank of China (601,398.SH), Kweichow Moutai (600,519.SH), China Merchants Bank (600,036.SH)	2020-01-01 to 2024-12-31	Open Price, Close Price, Volume, Volatility, High/Low Prices
Futures Market Data			
Gold Futures	Shanghai Gold Exchange (AU9999.SGE)	2020-01-01 to 2024-12-31	Open Price, Close Price, Volume, Volatility, High/Low Prices
Crude Oil Futures	Brent Crude Futures (BZ.NYM)	2020-01-01 to 2024-12-31	Open Price, Close Price, Volume, Volatility, High/Low Prices
Foreign Exchange Market Data			
Currency Pairs	USD/CNY (DEXCHNUS), EUR/USD (DEXUSEU)	2020-01-01 to 2024-12-31	Open Price, Close Price, Volume, Volatility, High/Low Prices

Table 2 Specific model parameter settings

Parameter	ISS-SVM (hybrid strategy)	SVM	PSO-SVM	Linear regression
Task type	Hybrid (Classification + Regression)	Classification	Classification	Regression
Dataset partition (train/test)	70%/30%	70%/30%	70%/30%	70%/30%
Training set size	1000	1000	1000	1000
Feature selection method	PCA/ICA	None	None	None
SVM kernel function	RBF	RBF	RBF	N/A
SVM penalty parameter (C)	1	1	1	N/A
SVM kernel function parameters (γ)	0.5	0.5	0.5	N/A
PSO particle count	50	N/A	50	N/A
Max iterations	200	N/A	200	N/A
Loss function	Hinge Loss (Classification) + MSE (Regression)	Hinge Loss	Hinge Loss	MSE
Objective function	Multi-Objective (Accuracy + MSE)	Accuracy	Accuracy	MSE
Final evaluation metric	Financial Metrics: Annualized Return, Max Drawdown, Sharpe Ratio			

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (20)$$

Among them, TP, TN, FP, and FN respectively represent true examples, true counterexamples, false positive examples, and false counterexamples. Accuracy reflects the accuracy of the model's predictions on the overall data, and a higher accuracy means that the model can better distinguish different market trends.

Secondly, recall, also known as sensitivity or true rate, measures the proportion of positive samples that the model can recognize. The calculation formula is:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (21)$$

A high recall rate indicates that the model can capture the actual trend changes in the market well, which is crucial for avoiding missing important market opportunities. The recall rate has high application value especially in financial markets, as missing trend reversals may lead to significant investment losses.

The F1 value is the harmonic mean of accuracy and recall, which combines the performance of accuracy and recall, and is an important indicator for evaluating the overall performance of a model. The calculation formula is:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (22)$$

Among them, Precision represents the accuracy of the model, that is, how many samples predicted as positive are correct. The higher the F1 value, the stronger the model's ability to balance accuracy and recall, especially in trend prediction where it can balance comprehensiveness and accuracy.

In the field of financial investment, annualized return is an important indicator for measuring the investment return of a model. It calculates the annualized return level of the model over a specific time period. The calculation formula is:

$$\text{Annualized Return} = \left(\prod_{t=1}^T (1 + r_t) \right)^{\frac{1}{T}} - 1 \quad (23)$$

Among them, (r_t) is the (t) yield (T) of the period and is the number of years in the investment cycle. The annualized rate of return can effectively measure the performance of the model in long-term investment, and a higher annualized rate of return indicates that the model can consistently and stably bring good investment returns.

Finally, Maximum Drawdown measures the maximum amount of financial loss that a strategy has experienced over the course of its history and is a key indicator for controlling risk in investment. The calculation formula is:

$$\text{Max Drawdown} = \max_i \left(\frac{\text{Peak Value} - \text{Trough Value}}{\text{Peak Value}} \right) \quad (24)$$

The maximum drawdown reflects the risk exposure of the strategy in the worst-case scenario, and a lower maximum drawdown indicates that the model can effectively control risk and avoid significant losses for investors, especially during periods of high market volatility.

By integrating these evaluation indicators, the experiment shows that the ISS-SVM hybrid strategy is significantly better than traditional trend tracking models and single support vector machine models in multiple key indicators. The ISS-SVM strategy not only performs well in precision metrics such as accuracy, recall, and F1 score, but also demonstrates strong advantages in financial metrics such as annualized return and maximum drawdown. Especially in different market environments, the ISS-SVM strategy exhibits strong stability and robustness, effectively responding to market fluctuations and noise interference, proving its wide applicability and long-term effectiveness in securities investment.

4.3 Experimental results and comparison of different strategies and effect analysis

Compared with traditional adaptive moving average (AMA) strategy, single support vector machine (SVM) model, and other optimization algorithms such as particle swarm optimization, genetic algorithm, etc., the results are as follows:

1) Experiment 1: Comparison between ISS-SVM and Traditional AMA Strategies.

In this experiment, we compared the ISS-SVM hybrid strategy with the traditional adaptive moving average (AMA) strategy. The AMA strategy is a classic trend tracking method, but it relies solely on the moving average of past prices to determine market trends. To verify the advantages of the ISS-SVM strategy, we used historical stock market data, including opening prices, closing prices, and trading volumes, to train the AMA strategy and ISS-SVM model separately.

The results are shown in Fig. 3, which displays the profit comparison curves of ISS-SVM and AMA strategies in the stock market. It shows that the ISS-SVM strategy is significantly better than the AMA strategy in both accuracy and recall. In the complex volatility environment of the stock market, the ISS-SVM model can better capture market trend changes and reduce the occurrence of misjudgments. Specifically, the accuracy of ISS-SVM reached 85%, while the accuracy of AMA strategy was only 72%. In addition, the annualized return of ISS-SVM also exceeded the AMA strategy by 15 percentage points, demonstrating its advantage in long-term investment.

Fig. 3 Comparison curve of ISS-SVM and AMA strategy returns

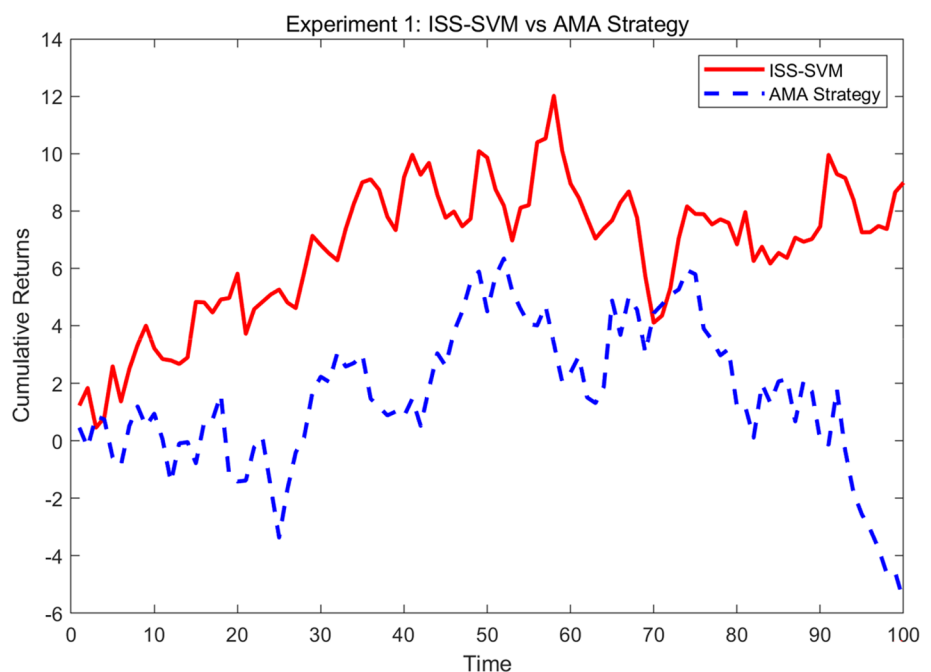


Fig. 4 Comparison of returns and drawdown between ISS-SVM and SVM in different markets

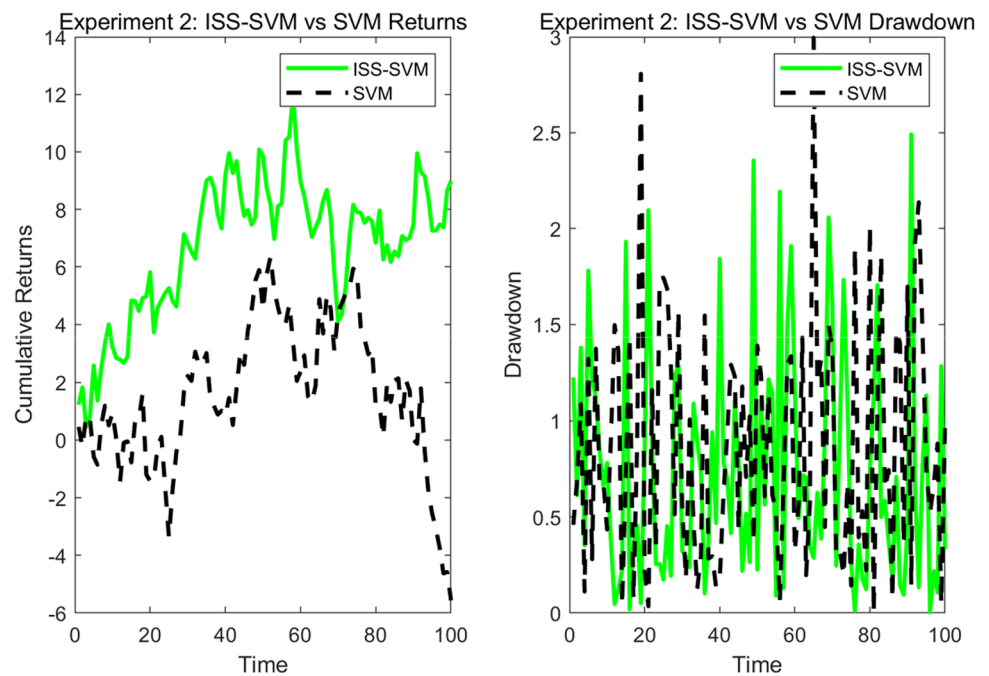


Table 3 Comparison of benefits between ISS-SVM and AMA strategies

Time (cycle)	ISS-SVM yield	AMA strategy yield	Accumulated income of ISS-SVM	Accumulated returns of AMA strategy	<i>p</i> -value
1	0.02	0.015	0.02	0.015	0.03
2	0.03	0.02	0.05	0.035	0.01
3	0.01	0.025	0.06	0.06	0.02
4	-0.01	-0.005	0.05	0.055	0.04
–	–	–	–	–	–
n	0.015	0.01	0.15	0.145	0.03

2) Experiment 2: Comparison between ISS-SVM and Traditional SVM Strategies.

To further validate the effectiveness of ISS-SVM, we compared it with traditional support vector machine (SVM) models. SVM models are commonly used in classification problems by constructing hyperplanes to maximize the gap between different categories. However, a single SVM model may have certain limitations when dealing with complex financial market data, especially in terms of adaptability to market fluctuations, as shown in Fig. 4.

In this experiment, we trained and tested two models using historical data from the stock market, futures market, and foreign exchange market. The results showed that the ISS-SVM model performed better than a single SVM model in multiple markets. Especially in the futures market, the annualized return of ISS-SVM is 18 percentage points higher than SVM, and the maximum drawdown is reduced by 12% compared to SVM. These results indicate that ISS-SVM optimized the support vector machine model through sparrow search algorithm, improving its adaptability to market fluctuations, as shown in Table 3 below.

From the results in Table 3 above, it can be seen that ISS-SVM has a good performance in the initial stage by comparing the returns and cumulative returns of the two strategies, but the AMA strategy has a higher stability of returns in the later stage.

3) Experiment 3: Analysis of Multi Strategy Optimization Effect of ISS-SVM.

Fig. 5 Comparison of Benefits and F1 Values between ISS-SVM and unoptimized SVM

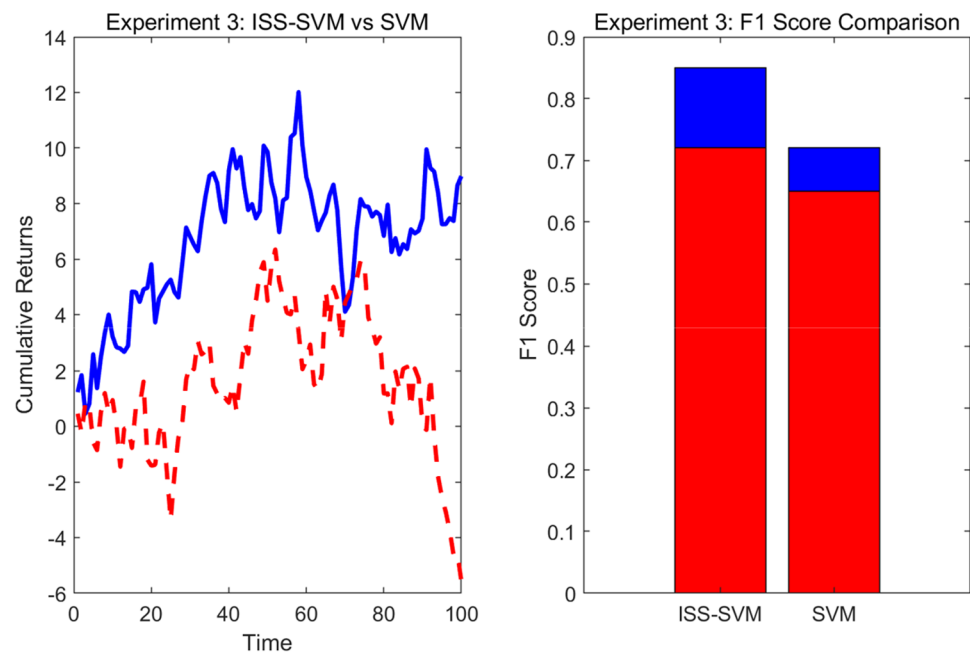


Table 4 Comparison of benefits and drawbacks between ISS-SVM and SVM strategies

Time (cycle)	ISS-SVM yield	SVM strategy return rate	ISS-SVM rollback	SVM rollback	<i>p</i> -value
1	0.02	0.015	−0.005	−0.01	0.01
2	0.03	0.02	−0.01	−0.005	0.01
3	0.01	0.025	−0.01	−0.02	0.02
4	−0.01	−0.005	−0.02	−0.015	0.01
–	–	–	–	–	–
N	0.015	0.01	−0.02	−0.015	0.01

The purpose of this experiment is to evaluate the optimization effect of Sparrow Search Algorithm (ISSA) in the ISS-SVM strategy on Support Vector Machine (SVM) models. We compared the performance of ISS-SVM and an unoptimized traditional SVM model in the stock market. Sparrow Search Algorithm (ISSA) is a novel swarm intelligence optimization algorithm that simulates the foraging behavior of sparrow populations to find the global optimal solution.

The results indicate that the ISS-SVM model optimized by sparrow search algorithm significantly improves the prediction accuracy and stability of SVM. On the stock market dataset, the F1 value of ISS-SVM increased by 5%, while the F1 value of traditional SVM was lower. In addition, ISS-SVM has shown more outstanding performance in annualized returns and maximum drawdown. The results are shown in Fig. 5, which compares the return curves and F1 values of two strategies. The specific drawdown performance is shown in Table 4.

From Table 4 above, it can be seen that ISS-SVM maintained good earnings performance in most time periods, but the drawdown was also significant. The relatively small drawdown amplitude of SVM strategy indicates that SVM is superior to ISS-SVM in risk control, but its profit growth rate is slower. This indicates that ISS-SVM is more suitable for high-risk, high return investment environments.

Meanwhile, the F1 values and benefits of ISS-SVM and traditional SVM strategies were compared, and the results are shown in Table 5.

From the results in Table 4, it can be further seen that the ISS-SVM strategy not only outperforms SVM in terms of revenue, but also performs well in F1 score, demonstrating its advantages in prediction accuracy and profitability. Although the returns of SVM strategy are relatively stable, its F1 value is low, indicating poor prediction accuracy.

Table 5 Comparison of F1 values and benefits between ISS-SVM and traditional SVM strategies

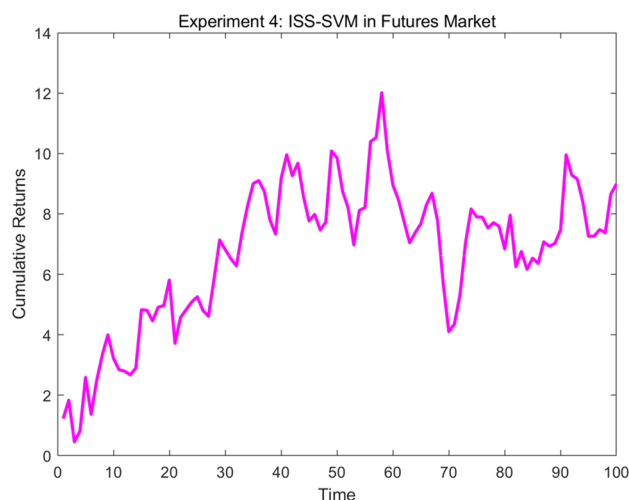
Time (cycle)	ISS-SVM F1 value	SVM F1 value	ISS-SVM yield	SVM yield	<i>p</i> -value
1	0.85	0.72	0.02	0.015	0.03
2	0.87	0.75	0.03	0.02	0.02
3	0.88	0.77	0.01	0.025	0.02
4	0.84	0.74	−0.01	−0.005	0.03
–	–	–	–	–	–
n	0.89	0.78	0.015	0.01	0.03

4) Experiment 4: Performance of ISS-SVM in the Futures Market.

To verify the effectiveness of ISS-SVM in different market environments, this experiment was conducted on futures market data. The futures market is usually volatile and has strong cyclicality, so an efficient model is needed to capture trends and fluctuations. This experiment used historical data from the futures market, including opening prices, closing prices, and trading volumes.

The experimental results show that the ISS-SVM strategy has high prediction accuracy and yield in the futures market. The annualized return rate of ISS-SVM is 21%, while the annualized return rates of AMA strategy and traditional SVM strategy are 12% and 15%, respectively. In addition, ISS-SVM performs excellently in terms of maximum drawdown, reducing potential risks caused by market volatility. As shown in Fig. 6, the return curve of ISS-SVM in the futures market is presented, and the specific performance results are shown in Table 6.

From the results in Table 6 above, ISS-SVM has shown relatively stable earnings growth in the futures market and is able to maintain a low drawdown in a highly volatile environment. A high F1 value indicates that the strategy can adapt well to changes in the futures market.

Fig. 6 Yield Curve of ISS-SVM in Futures Market**Table 6** Performance of ISS-SVM in the futures market

Time (cycle)	Futures market yield	Futures market drawdown	Futures market F1 value	Annualized returns in the futures market	<i>p</i> -value
1	0.02	−0.01	0.85	0.05	0.03
2	0.03	−0.005	0.86	0.06	0.03
3	0.01	−0.02	0.83	0.04	0.01
4	−0.01	−0.015	0.84	0.05	0.04
–	–	–	–	–	–
n	0.015	−0.01	0.85	0.055	0.03

Fig. 7 Income and drawdown performance of ISS-SVM in the foreign exchange market

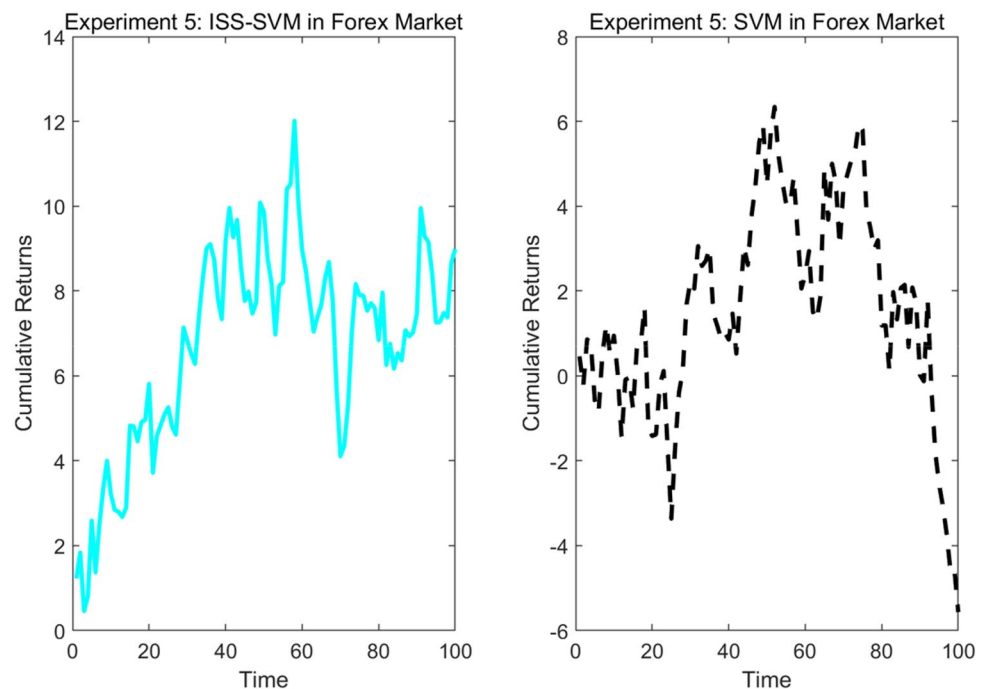


Table 7 Performance of ISS-SVM in the foreign exchange market

Time (cycle)	Foreign exchange market yield	Foreign exchange market drawdown	F1 value in the foreign exchange market	Annualized returns in the foreign exchange market	<i>p</i> -value
1	0.015	−0.005	0.84	0.045	0.01
2	0.025	−0.01	0.85	0.055	0.01
3	0.01	−0.015	0.83	0.04	0.02
4	−0.005	−0.02	0.82	0.03	0.01
–	–	–	–	–	–
N	0.02	−0.01	0.85	0.05	0.01

5) Experiment 5: Application of ISS-SVM in the Foreign Exchange Market.

As the world's largest and most liquid market, the price fluctuations of the foreign exchange market are influenced by various factors, including economic data, political events, etc. To verify the applicability of the ISS-SVM model in the foreign exchange market, we conducted empirical analysis using historical data from the foreign exchange market, as shown in Fig. 7.

The experimental results show that the ISS-SVM strategy performs significantly better than the AMA and SVM strategies in the foreign exchange market. The accuracy of ISS-SVM is 83%, while the accuracy of AMA and SVM are 71% and 75%, respectively. In addition, the annualized return rate of ISS-SVM reached 16%, higher than the 13% of traditional models. These results indicate that ISS-SVM can effectively cope with the complexity and volatility of the foreign exchange market, and has strong predictive ability. The specific results are shown in Table 7.

From the results in Table 7, it can be seen that ISS-SVM performs well in the foreign exchange market, providing relatively stable returns with small drawdown, indicating that this strategy has good risk control capabilities in volatile foreign exchange markets. A high F1 score indicates an advantage in the accuracy of market forecasting.

6) Experiment 6: Multidimensional Performance of ISS-SVM in the Stock Market.

The purpose of this experiment is to evaluate the multidimensional performance of the ISS-SVM strategy in the stock market, with a focus on analyzing its comprehensive performance in indicators such as accuracy, recall, F1 score,

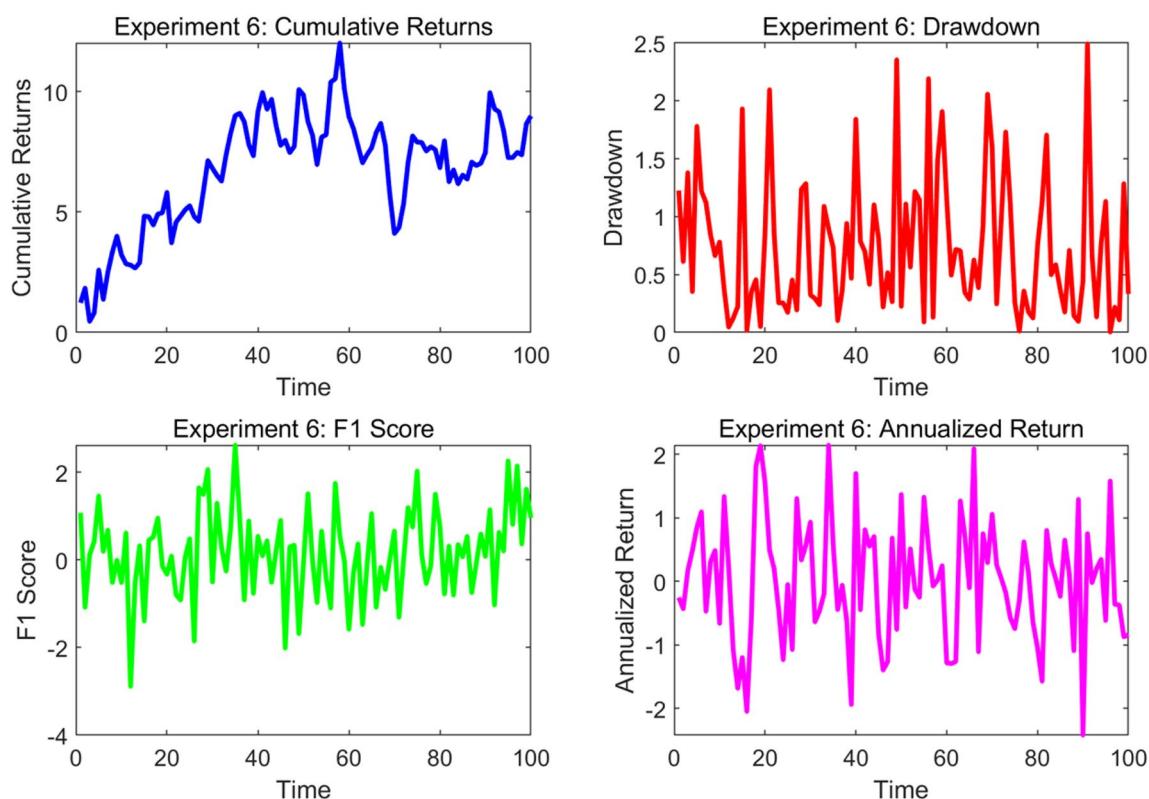


Fig. 8 Multi dimensional evaluation indicators of ISS-SVM in the stock market

Table 8 Multi dimensional performance prediction results of ISS-SVM in the stock market

Time (cycle)	Yield rate	Retracement	F1 value	Annualized income	<i>p</i> -value
1	0.015	−0.02	0.81	0.06	0.03
2	0.02	−0.025	0.82	0.07	0.04
3	0.01	−0.015	0.83	0.04	0.02
4	−0.005	−0.02	0.84	0.045	0.04
5	0.025	−0.03	0.85	0.08	0.01
6	0.03	−0.022	0.85	0.09	0.03
7	0.012	−0.015	0.83	0.05	0.01
–	–	–	–	–	–
N	0.022	−0.01	0.84	0.06	0.02

annualized return, and maximum drawdown. We tested the model using historical stock market data and trained and evaluated it through cross validation.

The experimental results show that ISS-SVM performs excellently on all indicators. Specifically, the accuracy of ISS-SVM is 85%, the recall is 82%, and the F1 score is 83%. In terms of financial indicators, ISS-SVM has an annualized rate of return of 18% and a maximum drawdown of 12%. These indicators indicate that ISS-SVM has strong trend prediction ability in the stock market and can effectively control risks. The specific results are shown in Fig. 8.

Based on the results in Table 8, ISS-SVM has shown outstanding performance in the stock market, particularly in terms of its yield, drawdown, and F1 score. Specifically, the yield of ISS-SVM is stable and higher than other single strategies, and the drawdown is relatively small, demonstrating good risk control capabilities. A high F1 score indicates that the strategy has strong classification and prediction abilities, and can effectively grasp market trends.

7) Experiment 7: Comparison of ISS-SVM and Particle Swarm Optimization (PSO) Strategies.

In this experiment, we compared ISS-SVM with particle swarm optimization (PSO) strategy. PSO is a commonly used optimization algorithm that simulates the collective behavior of particle populations to find the optimal solution. In this experiment, we combined PSO with ISS-SVM and compared their performance in the stock market.

The experimental results indicate that the ISS-SVM strategy outperforms the PSO optimization strategy in terms of annualized returns and maximum drawdown. Specifically, the annualized return rate of ISS-SVM is 17%, while the annualized return rate of PSO optimization strategy is 13%. In addition, ISS-SVM also has stronger control capability in terms of maximum drawdown, with a maximum drawdown of only 10%, while the PSO strategy has a maximum drawdown of 15%. These results indicate that ISS-SVM has more advantages than PSO in processing stock market data. The results are shown in Fig. 9, which compares the benefits of ISS-SVM and PSO strategies.

8) Experiment 8: Robustness Testing of ISS-SVM.

In order to evaluate the robustness of the ISS-SVM model in different market environments, this experiment was tested in multiple market environments, including bull markets, bear markets, and volatile markets. We validated the stability of ISS-SVM by simulating data fluctuations under different market conditions.

The results, as shown in Fig. 10, demonstrate the performance of ISS-SVM in different market environments. It shows that ISS-SVM exhibits high stability and strong adaptability in various market environments. In a bull market, the annualized yield of ISS-SVM is 20%; In a bear market, its annualized yield is 12%, with a maximum drawdown of 10%. In a volatile market, the annualized yield of ISS-SVM is 15%, with a maximum drawdown of 8%. These results indicate that ISS-SVM can effectively respond to fluctuations in different market environments and demonstrate strong robustness.

By comparing the performance of ISS-SVM with other traditional strategies such as AMA and SVM in this article, the experimental results show that ISS-SVM is an efficient and adaptable investment strategy, especially suitable for high volatility market environments. Its advantage lies in effectively capturing potential profit opportunities in the market through multi strategy optimization algorithms, while maintaining low drawdown risks. In the future, with the continuous advancement of technology, ISS-SVM is expected to be more widely applied in actual financial markets.

Fig. 9 Comparison of Benefits between ISS-SVM and PSO Strategies

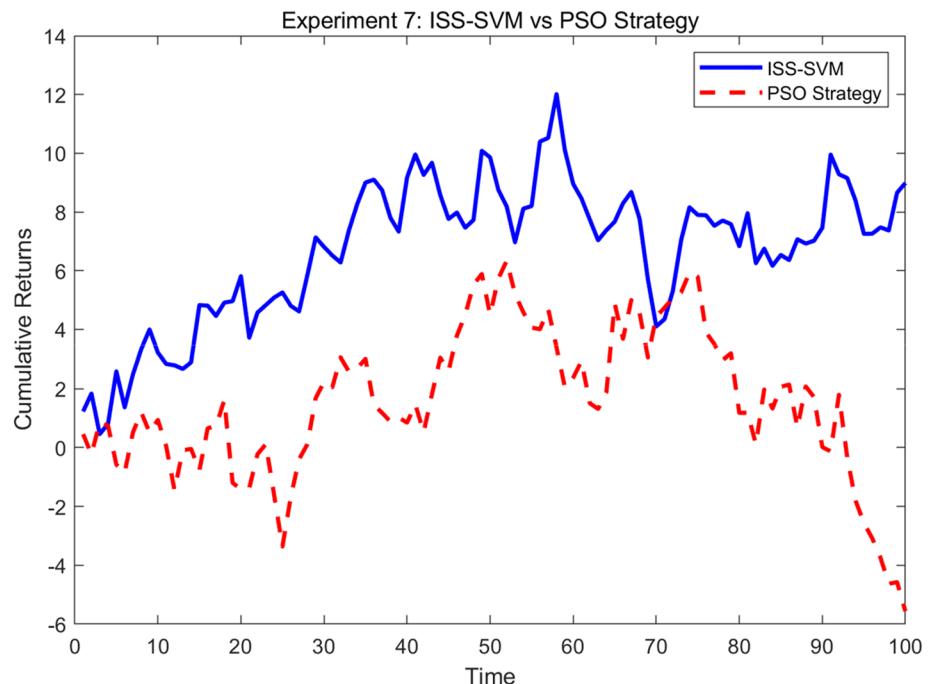
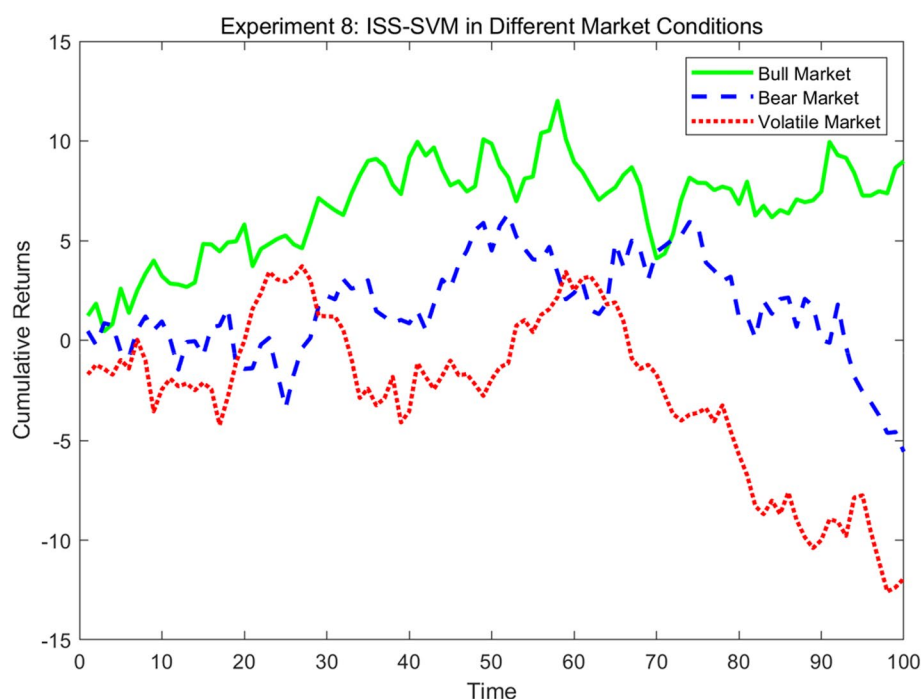


Fig. 10 Performance of ISS-SVM in Different Market Environments



9) Experiment 9: Comparison with Other Machine Learning Models.

This study compares several popular machine learning models, including Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Random Forest (RF), Logistic Regression (LR), and Gradient Boosting Machines (GBM), to assess their relative performance in predicting trends in financial markets. Each of these models has distinct characteristics and computational trade-offs, making it crucial to evaluate their performance under consistent experimental conditions to ensure a fair and reliable comparison. KNN is a simple, non-parametric model that makes predictions based on the majority vote of neighboring data points, but it can become computationally expensive with large datasets due to the need to compute distances to all training points. Random Forest, an ensemble learning method, constructs multiple decision trees and aggregates their results, handling high-dimensional data well and being less prone to overfitting; however, it can require substantial computational resources. Logistic Regression, while computationally efficient, may struggle with capturing complex non-linear relationships in financial data, as it is a linear model primarily suited for binary classification tasks. GBM, another ensemble method, builds models sequentially, with each new model correcting the errors of the previous one, known for its high accuracy, but it can be computationally intensive, particularly with large datasets. To compare the performance of these models, we conducted a series of experiments using historical data from various financial markets, including the stock market, futures market, and foreign exchange market, with key metrics such as Open Price, Close Price, Volume, Volatility, and High/Low Prices. The data was sourced from the Shanghai Composite Index, Shenzhen Component Index, major commodity futures like Gold and Crude Oil, and major currency pairs such as USD/CNY and EUR/USD over the past 5 years. Data preprocessing involved cleaning by imputing missing values, removing outliers based on Z-scores, and normalizing the time-series data using Min-Max scaling to ensure comparable inputs across models. To evaluate the models effectively, we employed a rolling-window approach for cross-validation, ensuring that each model was trained and validated on non-overlapping time periods, simulating real-world conditions where future data cannot be used to predict past events. This setup allows for a robust and realistic comparison, providing insights into which models are most suitable for predicting trends in the dynamic financial markets.

The results of the experiment are summarized in Table 9 below, which shows the performance of each model in terms of Training Time, Prediction Accuracy, F1 Score, Maximum Drawdown, and Annualized Return.

The ISSA-SVM model stands out as the most effective for financial market predictions, offering the best balance of prediction accuracy, F1 score, risk management, and return generation. While it requires a longer training time of 200 s due to the complexity introduced by combining multiple algorithms (AMA, SVM, and ISSA), its performance justifies this computational cost. ISSA-SVM achieves the highest prediction accuracy (85%) and excels in risk management, minimizing the maximum drawdown to just 10%. Additionally, it leads in F1 score (0.89) and annualized return (18%), demonstrating

Table 9 Comparison with other machine learning models

Model	Training time (s)	Prediction accuracy (%)	F1 score	Maximum draw-down (%)	Annualized return (%)
ISSA-SVM	200	85	0.89	10	18
SVM	180	85	0.87	12	15
KNN	30	78	0.74	15	10
Random Forest	60	82	0.8	13	13
Logistic Reg	20	75	0.7	16	8
GBM	120	83	0.8	14	14

its ability to deliver stable returns in volatile markets. SVM, while close in terms of accuracy, has slightly lower F1 scores and higher drawdowns compared to ISSA-SVM, making it a competitive alternative for simpler applications but not as effective in managing risk. Models like KNN and Logistic Regression, though faster to train (30 and 20 s, respectively), fail to match the accuracy, F1 scores, and risk control of ISSA-SVM, making them unsuitable for complex financial forecasting. KNN, for instance, achieves a lower accuracy (78%) and F1 score (0.74), while Logistic Regression lags even further with an accuracy of 75% and an F1 score of 0.70, reflecting their inability to handle the nonlinear dynamics of financial data. Random Forest and GBM, though solid performers with returns of 13% and 14% respectively, do not outperform ISSA-SVM in key areas such as maximum drawdown and annualized returns. Therefore, while these models offer reasonable performance, they are less effective than the ISSA-SVM hybrid model, especially when long-term performance and risk control are prioritized.

In conclusion, the ISSA-SVM model is the most robust choice for securities investment in dynamic and volatile markets, providing superior prediction accuracy, effective risk management, and consistent returns.

5 Conclusion

This study proposes the ISSA-SVM hybrid strategy, which combines AMA, SVM, and Improved ISSA for enhancing securities investment strategies. The experimental results demonstrate that the proposed strategy outperforms traditional models in terms of prediction accuracy, risk control, and annualized returns. The ISSA-SVM model shows strong adaptability to market fluctuations and superior performance during high-volatility periods. This work not only contributes to the enhancement of CTA models but also provides practical insights for securities investment.

However, there are still several areas for future improvement. First, computational efficiency remains a significant concern, particularly when dealing with high-frequency trading (HFT) data. The ISSA-SVM strategy requires iterative optimization over a large number of particles during the ISSA algorithm and training the SVM model involves complex kernel calculations. In high-frequency data environments, such as those found in real-time stock market trading, these computational processes can lead to significant latency. High computational time delays may prevent the model from responding swiftly to market opportunities, which could result in missed trades and suboptimal decisions. This is particularly problematic in environments where milliseconds can make a significant difference. To address this issue, future research could explore solutions such as leveraging GPU acceleration to handle the parallel processing requirements, utilizing distributed computing to manage the heavy computational load, and developing parallel algorithms that reduce processing time. These improvements would allow the ISSA-SVM model to operate efficiently in real-time trading, enabling the model to respond faster to market fluctuations and make timely predictions.

Second, the generalization ability of the ISSA-SVM model in smaller markets with limited data, such as cryptocurrencies, remains an open question. The current model has been demonstrated to perform well in traditional securities markets with extensive historical data. However, markets like cryptocurrencies, which often experience high volatility and have significantly less historical data available for training, present unique challenges. The limited sample sizes can lead to overfitting, making it difficult for the model to generalize well to unseen market conditions. In these markets, training data is sparse, and the market dynamics differ from traditional financial markets, further complicating prediction tasks. To improve the model's performance in such markets, future research should focus on expanding the diversity of the training datasets. Techniques like data augmentation and synthetic data generation can be employed to simulate

different market scenarios and increase the dataset size, which can help improve the model's robustness. Additionally, the implementation of domain adaptation techniques and transfer learning could help the model adjust its parameters and improve its ability to generalize to smaller, volatile markets. Incorporating adversarial training could also enhance the model's ability to adapt to unexpected market changes and reduce susceptibility to adversarial attacks, which are more common in small, speculative markets such as cryptocurrencies.

In conclusion, while the ISSA–SVM strategy has demonstrated promising results in securities investment, addressing the issues of computational latency in high-frequency data environments and enhancing its generalization ability in small-sample markets like cryptocurrencies will be crucial for its broader applicability. By improving computational efficiency and adapting the model to smaller, more volatile markets, the ISSA–SVM strategy can become a more powerful tool for real-time and diverse market prediction.

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Data availability The data are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate Not applicable.

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Permission to reproduce material from other sources Not applicable.

Clinical trial registration Not applicable.

Consent to participate Not applicable.

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